# SaiLER

*Sparse Switching and Sharing for Lifelong Learning with Reinforcement*

DARPA ShELL

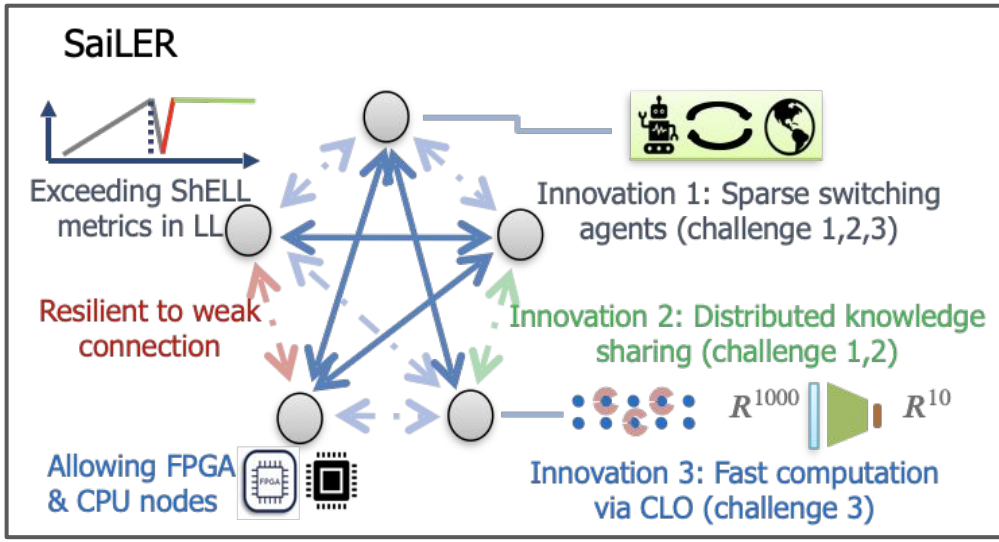
JOHNS HOPKINS UNIVERSITY

DARPA

UCLA

# Overview of SaiLER

JOHNS HOPKINS UNIVERSITY UCLA




## Innovations to Overcome ShELL Challenges:

- Minimizing what to share using novel *sparse switching cost RL*
- Using *distributed database and protocols to linearly faster knowledge sharing*
- Reducing the memory and compute using *Compressive Linear Operators (CLO)* for SWAP requirements

## Team members:

- JHU: Vladimir Braverman, Michael Jacobs, Vishwa Parekh, Jingfeng Wu, Zhuolong Yu, Aditya Krishnan, Ziang Song, Michael Hu, Haoran Li, Shuhao Lai
- UCLA: Lin F. Yang, Khushbu Pahwa, Jialin Dong

## Impact and Deliverables:

- SaiLER will deliver two systems: CANAL and MARINE
- CANAL: Proof of concept. SWAP devices for Atari
- MARINE: Main deliverable. ShELL capabilities for medical imaging in the field:
  - Mobile hospitals, fleets, war zones

# CANAL: Continual Atari games with Networking Agent Learning

- Deploy RL agents to learn Atari games on low-power devices (LPD)
  - PYNQ, Nvidia Jetson, ...
  - Star topology
  - Open AI Gym
  - Low-switching cost RL
- Achieve lifelong learning ability with L2M algorithms including memory replay and EWC
- Distributed aggregative knowledge to achieve shared-experience learning across agents
- Reduce computation and communication cost with Compressive Linear Operators (CLO) while preserving learning performance

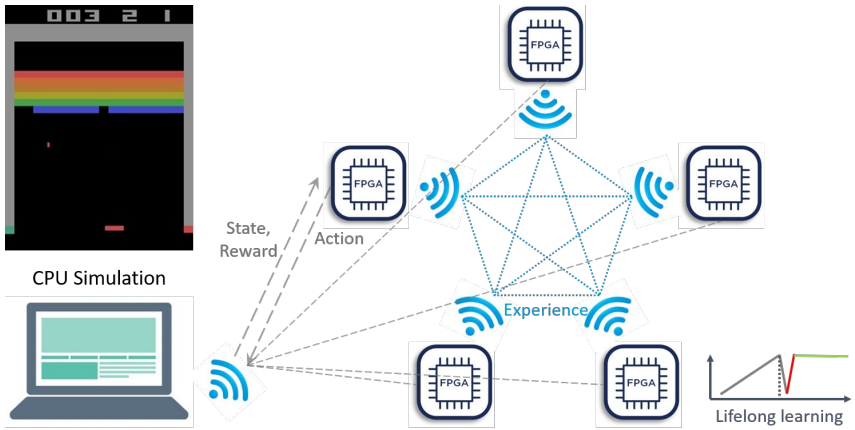
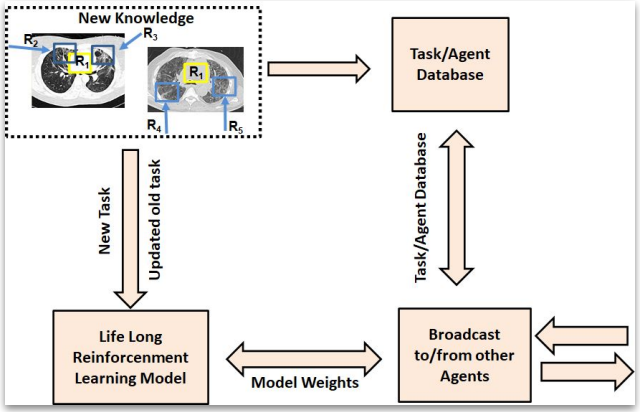
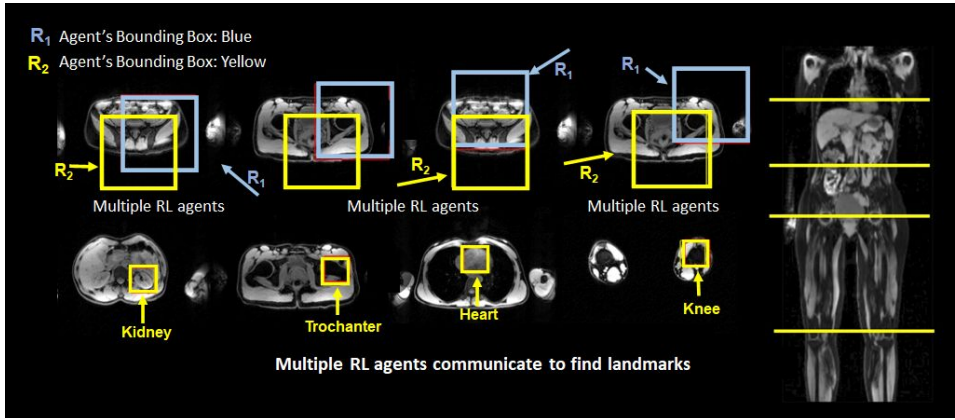



Figure 4: CANAL System

# **MARINE:** Medical image Analysis and Reasoning with Interactive NEtworks

- Localize a set of anatomical landmarks (e.g. heart, COVID-19 lung patterns, etc.) using multiagent lifelong distributed RL models.
- Implement shared knowledge representation across different nodes to enable ShELL learning of new (eg. new anatomical landmark: knee) and existing tasks (eg. novel unseen COVID-19 lung patterns)
- Implement L2M algorithms including memory replay and EWC to achieve task-incremental lifelong learning
- SWAP computing nodes: standard PCs, laptops

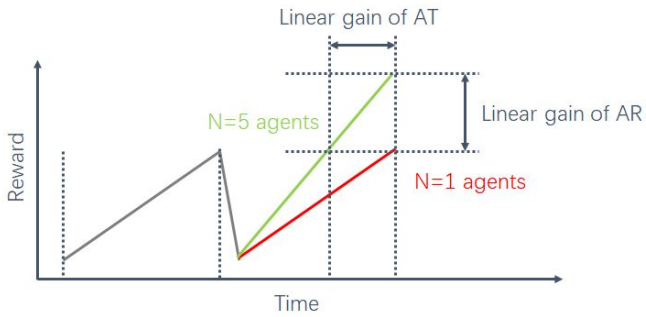


# Metrics

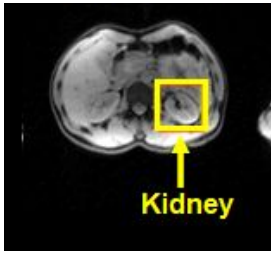
AT (Agent Time):

- Average time of N=5 agents achieving average reward alpha over K=5 distinct sequential tasks
  - CANAL: alpha = 90% (average **normalized scores** for all tasks)
  - MARINE: alpha = agents are **at most 5 mm away from the centers of the organ bounding boxes**
  - Each agent learn tasks in different sequence with shared knowledge




AR (Agent Reward):

- Average reward of N=5 agents after training a fixed time T on each of the K=5 distinct sequential tasks
  - CANAL: reward = normalized scores for each task
  - MARINE: reward = distance between agent and the center of the organ bounding box




Additional metrics:

  - CL metrics: forward transfer, backward transfer
  - SWAP metrics: model size, memory cost, power usage

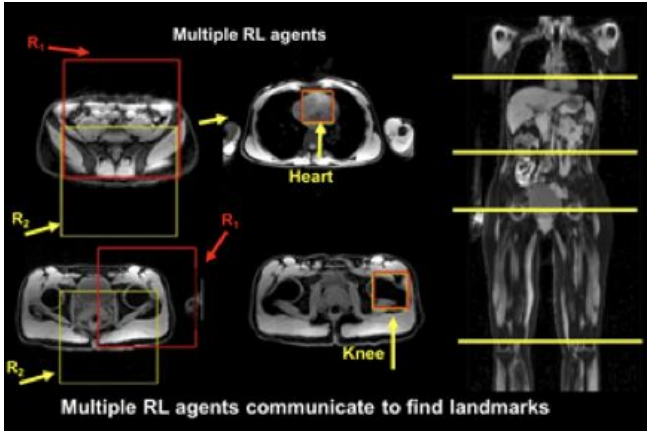
# Milestones

**PHASE 1**

Milestone 1 (M2) Develop novel RL framework for low switching cost
Milestone 2 (M4) Design SaiLER, CANAL and MARINE system
Milestone 3 (M6) Initial demonstration of simulated CANAL system

**PHASE 2**

Milestone 1 (M9) Evaluation on CANAL system for 5 agents on LPD.
Milestone 2 (M11) Refine novel communication protocols for sharing
Milestone 3 (M13) Initial demonstration of MARINE system
Milestone 4 (M15) Refine and extend evaluation on MARINE system
Milestone 5 (M18) Final demonstration, report and open-source code



# CANAL

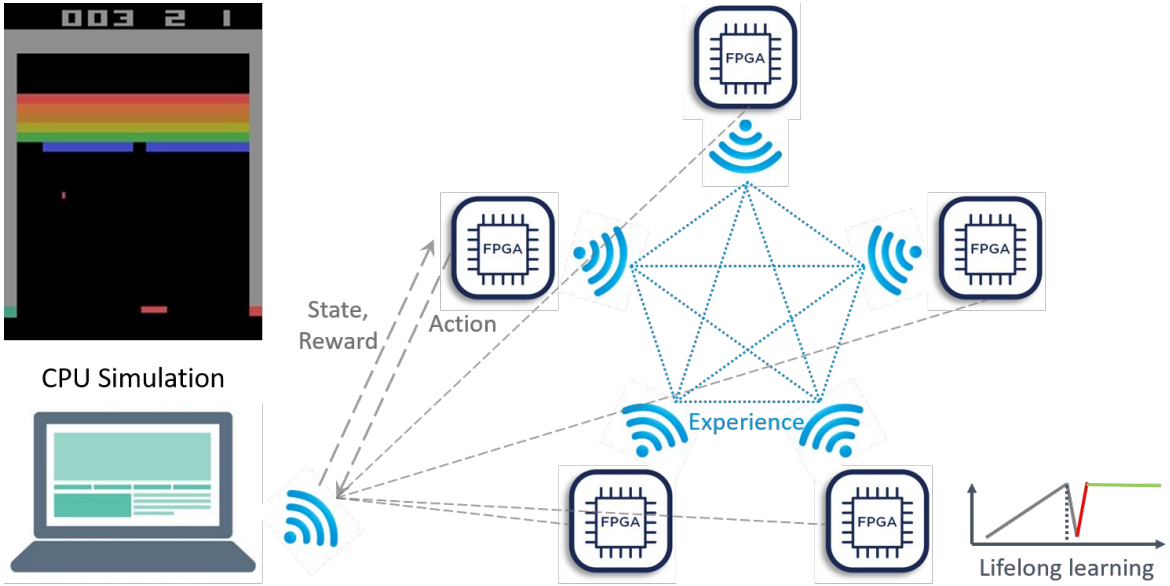
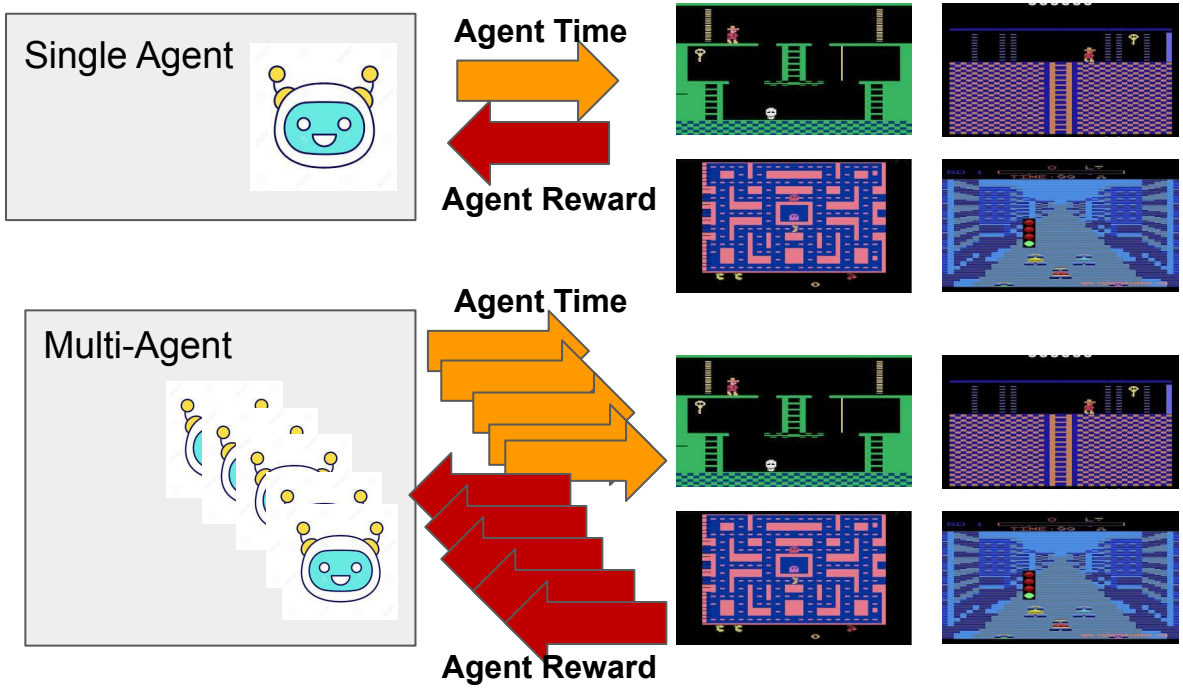



Figure 4: CANAL System

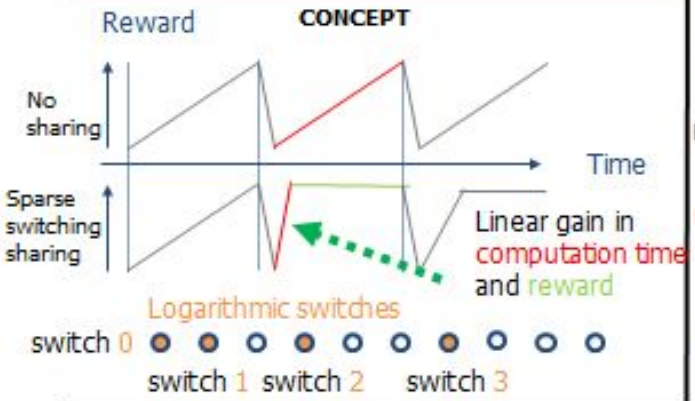

Single Agent
Agent Time
Agent Reward
Multi-Agent
Agent Time
Agent Reward

# SaiLER: Sparse Switching and Sharing for Lifelong Learning with Reinforcement

JHU; Technical POC : Vladimir Braverman (sub UCLA)

DARPA-PA-20-02
Artificial Intelligence Exploration (AIE)

## CONCEPT




## APPROACH

## IMPACT

1. SHELL capabilities for medical imaging in the field:
   - Mobile hospitals, fleets, war zones
2. Deployable with any lifelong LL/RL agent (e.g. PPO)
3. Low-cost integrated response to multiple threats in the field

## CONTEXT

1. Existing LL is often expensive to share, e.g., FPGA
2. Existing LL is too expensive for low-SWAP devices.
3. Existing LL sharing scales with the actual model size, instead of the intrinsic size
4. No LL systems with low switching cost sharing
5. No LL system for medical imaging

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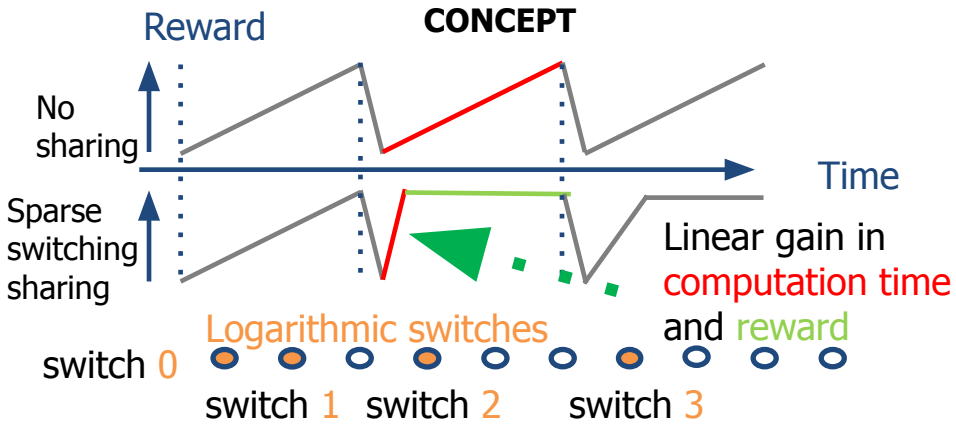


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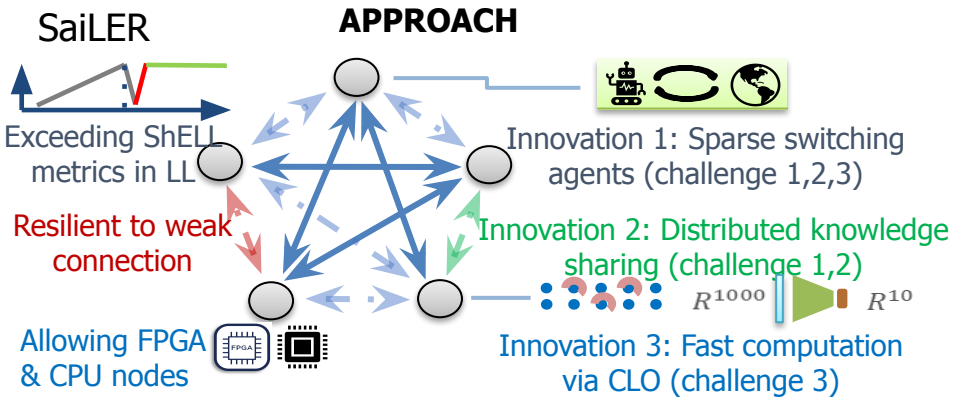


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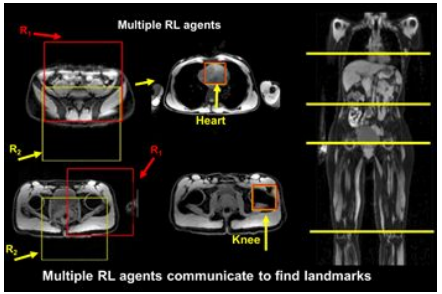


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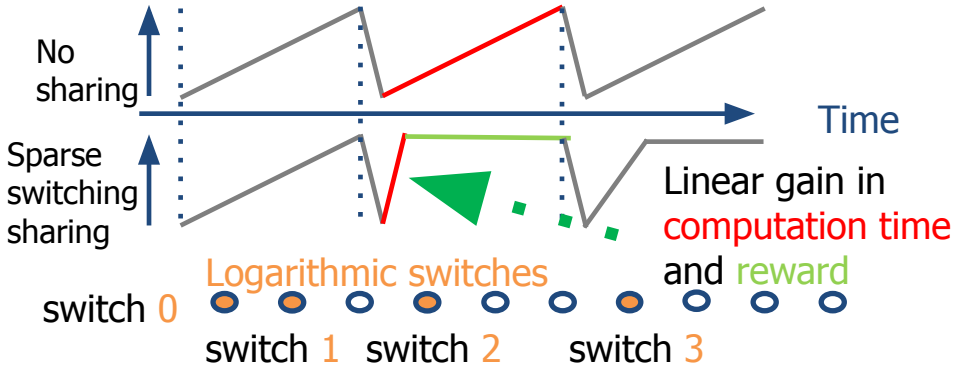


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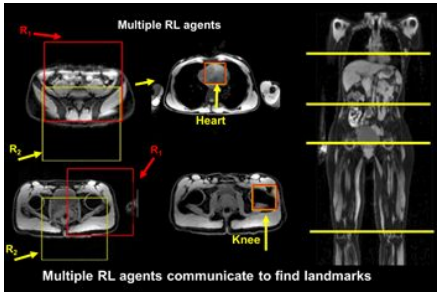
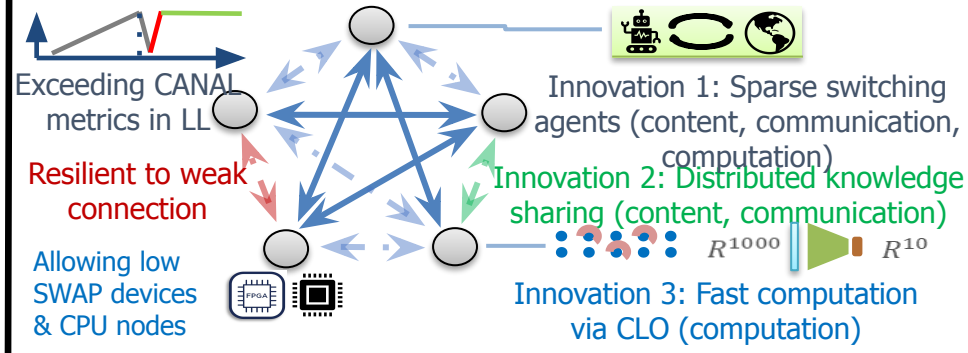



## CONTEXT

### Overview

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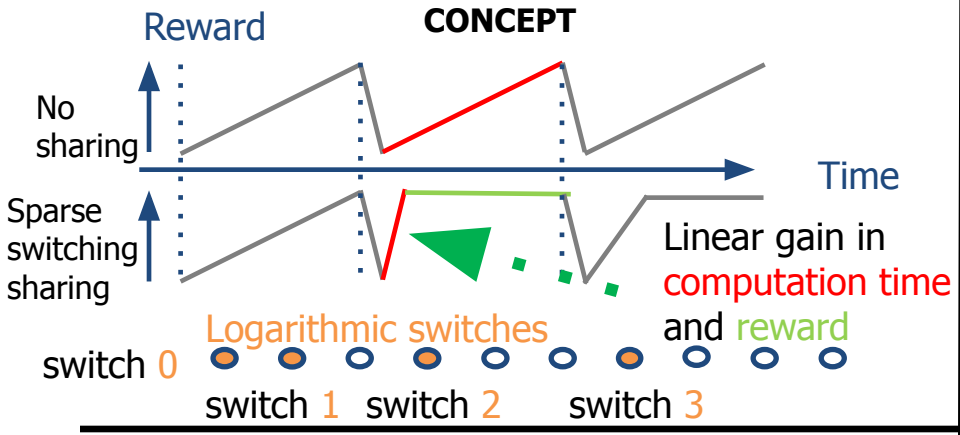
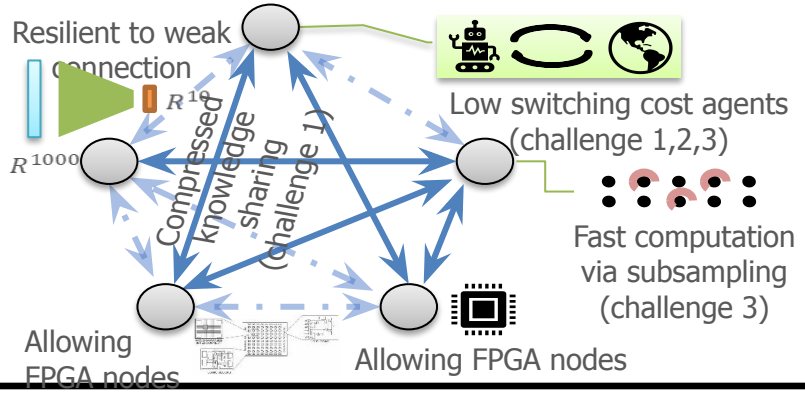



## SaiLER APPROACH

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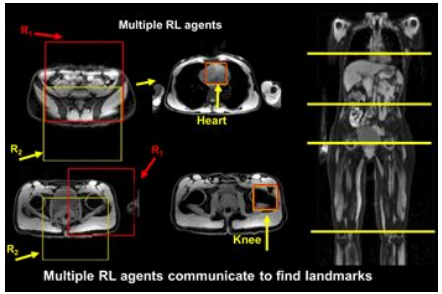


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Organization Name(s); Technical POC Name(s)



**CONCEPT**

Provide graphic.

**APPROACH**

Describe new ideas.

**IMPACT**

Describe need and problem being addressed.

Describe goal.

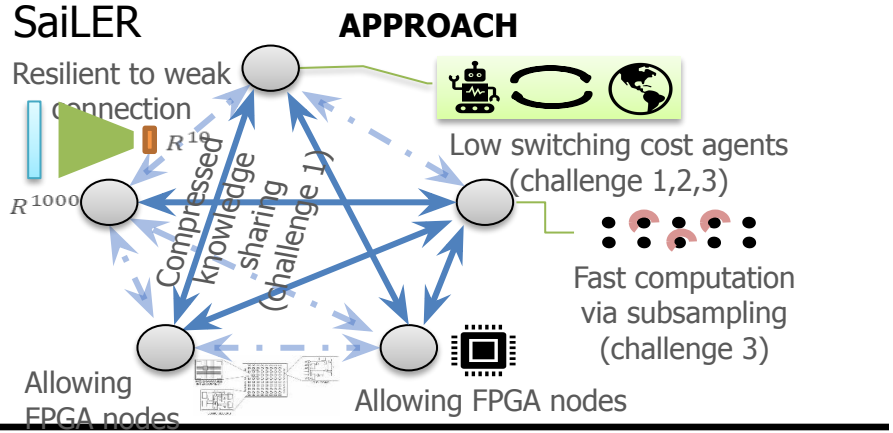
**CONTEXT**

Describe existing approaches/state of the art.

**CONCEPT**

**IMPACT**

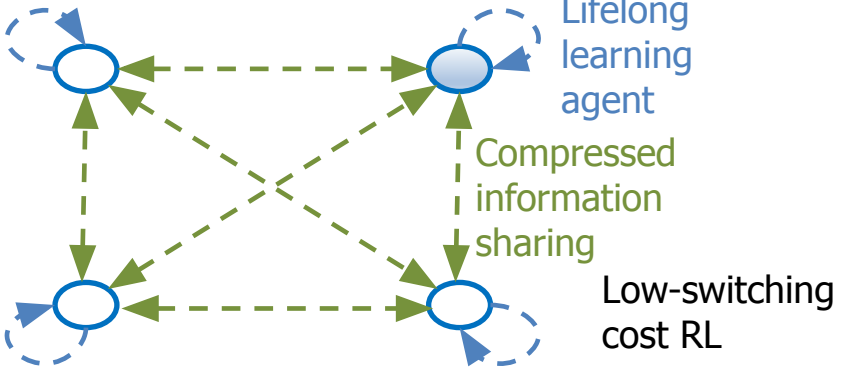
**APPROACH**




**CONTEXT**

**CONCEPT**

**APPROACH**




**IMPACT**

**CONTEXT**

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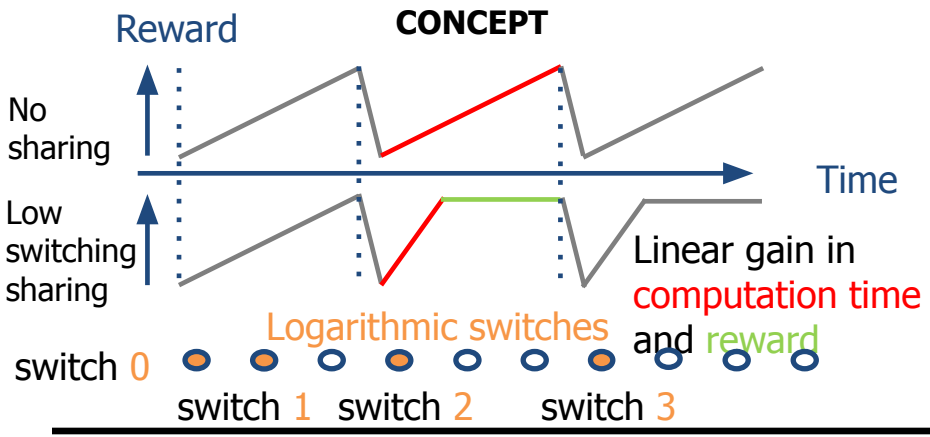


## APPROACH

## IMPACT

## CONTEXT

# **MARINE:** Medical image Analysis and Reasoning with Interactive NEtworks

- Goal: localize a set of anatomical landmarks (e.g. heart, COVID-19 lung patterns, etc.) using multiagent lifelong distributed RL models.

- Implement shared knowledge representation across different nodes to enable ShELL learning of new (eg. new anatomical landmark: knee) and existing tasks (eg. novel unseen COVID-19 lung patterns)

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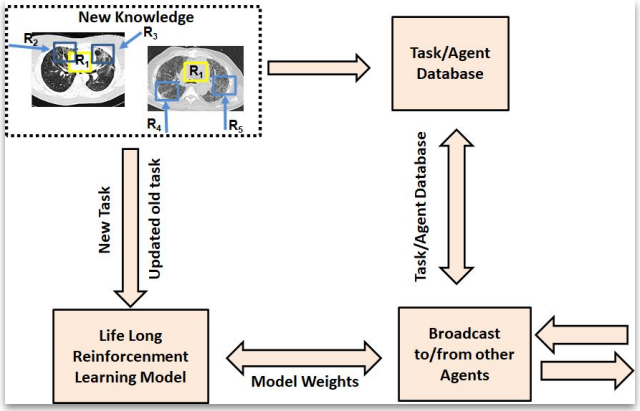
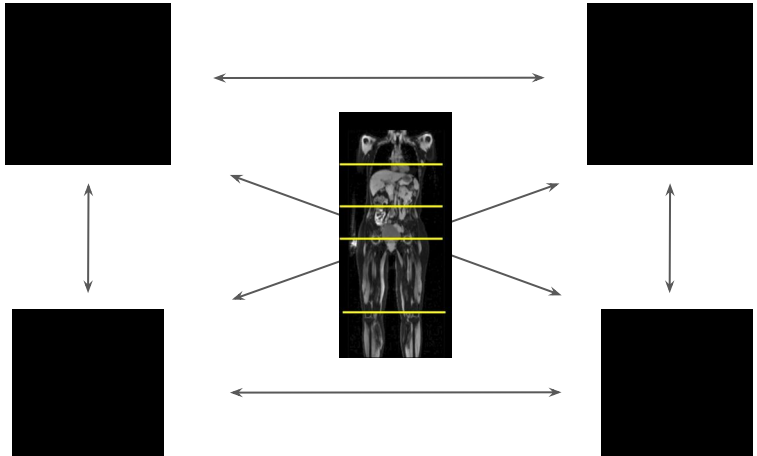
- SWAP computing nodes: standard PCs, laptops



# Full Proposal Title

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## CONCEPT

Provide graphic.

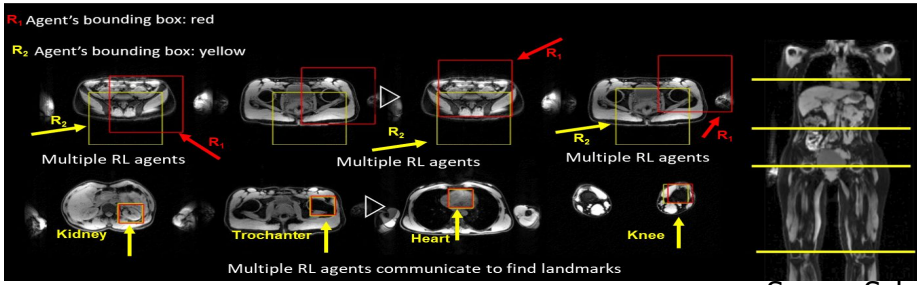
## APPROACH

Describe new ideas.

## IMPACT

Describe need and problem being addressed.

Describe goal.

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Provide graphic.

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Describe new ideas.

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Describe need and problem being addressed.

Describe goal.

**CONTEXT**

Describe existing approaches/state of the art.

# Additional Metrics for SWAP

- To demonstrate superiority of SaiLER under SWAP requirements, we investigate efficiency and applicability under the SWAP context
- We are developing more metrics to investigate system performance under SWAP
- We are also evaluating feasibility / cost of deploying our CL solutions on SWAP hardware. Options include but are not limited to:
    - Raspberry Pi
    - PYNQ
    - Nvidia Jetson series
- Hardware metrics e.g. model complexity, memory cost, inference throughput will also be measured against simulated and realistic use cases

# Milestones and Tasks

Someone please copy a few milestones from the proposal

SHould we do in Phases?

PHASE 1
Milestone 1  Develop novel RL framework for low switching cost
Milestone 2  Preliminary  Research Plan of SaiLER, CANAL and MARINE: Design report:
Milestone 3  Initial Demonstration of CANAL with SaiLER: Design report

PHASE 2
Milestone 1  Develop LPD-based evaluation of the CANAL system for 5 agents.
Milestone 2  Develop and refine novel communication protocols for sharing of knowledge.
Milestone 3  Initial Demonstration of Marine System with SaiLER
Milestone 4  Build and run basic benchmarks

# Metrics

- AT(AGENT-TIME): Computational time of agent when solving a new task
- AR(AGENT-REWARD): Reward of a single agent solving a new task after a fixed number of training episodes
- Additional metrics for measuring communication, memory, power and accuracy

# Metrics

AR (Agent Reward):

- Baseline: Average reward of a single agent after training fixed time T on each of the K sequential tasks.
  - CANAL: K = 5, reward = normalized scores for all tasks
  - MARINE: K = 5, reward = distance between agent and the center of the organ bounding box
- Target: N=5 agents, average reward of agents after training fixed time T on each of the K sequential tasks
  - K=5 distinct task for all agents
  - Agent may learn tasks in different sequence

https://arxiv.org/pdf/1509.06461v3.pdf

# Metrics

CL metrics:

- Forward transfer
- Backward transfer

Communication, memory etc.

# Metrics for MARINE

AT (Agent Time):

- Baseline: Time of a single agent to achieve average reward at least alpha over k sequential tasks.
  - K = 5, alpha = agents are at most 5 mm away from the centers of the organ bounding boxes

- Target: N=5 agents, average time of agents achieving average reward at least alpha over k sequential tasks
  - k=5 distinct organs for all agents
  - Agents may start from different organs

# Metrics for CANAL

score for each game as follows:

$$\text{score}_{\text{normalized}} = \frac{\text{score}_{\text{agent}} - \text{score}_{\text{random}}}{\text{score}_{\text{human}} - \text{score}_{\text{random}}}.$$

AT (Agent Time):

- Baseline: Time of a single agent to achieve average reward at least alpha over k sequential tasks.
  - K = 5, alpha = 90% average normalized scores for all tasks

- Target: N=5 agents, average time of agents achieving average reward at least alpha over k sequential tasks
  - k=5 distinct Atari-games for all agents
  - Agents may start from different games

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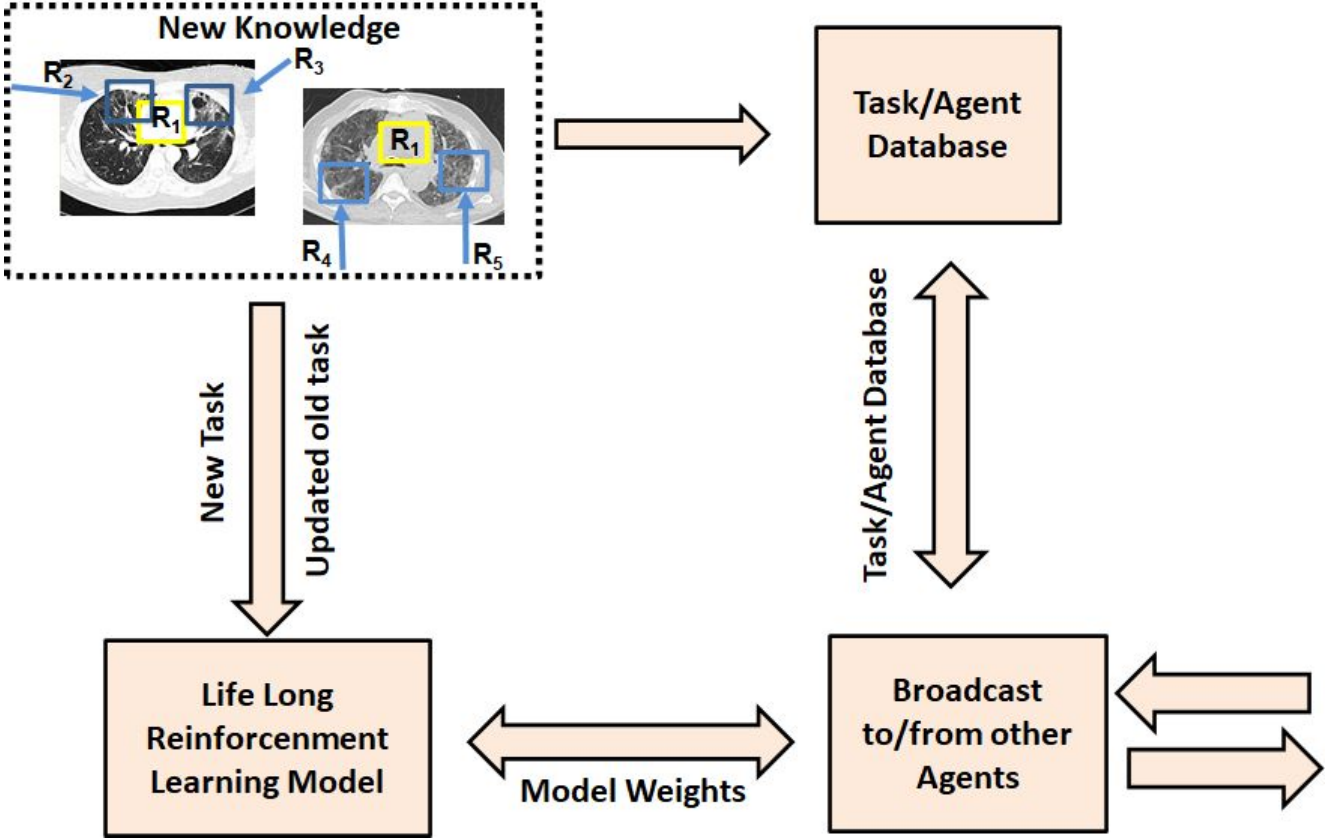
# Details

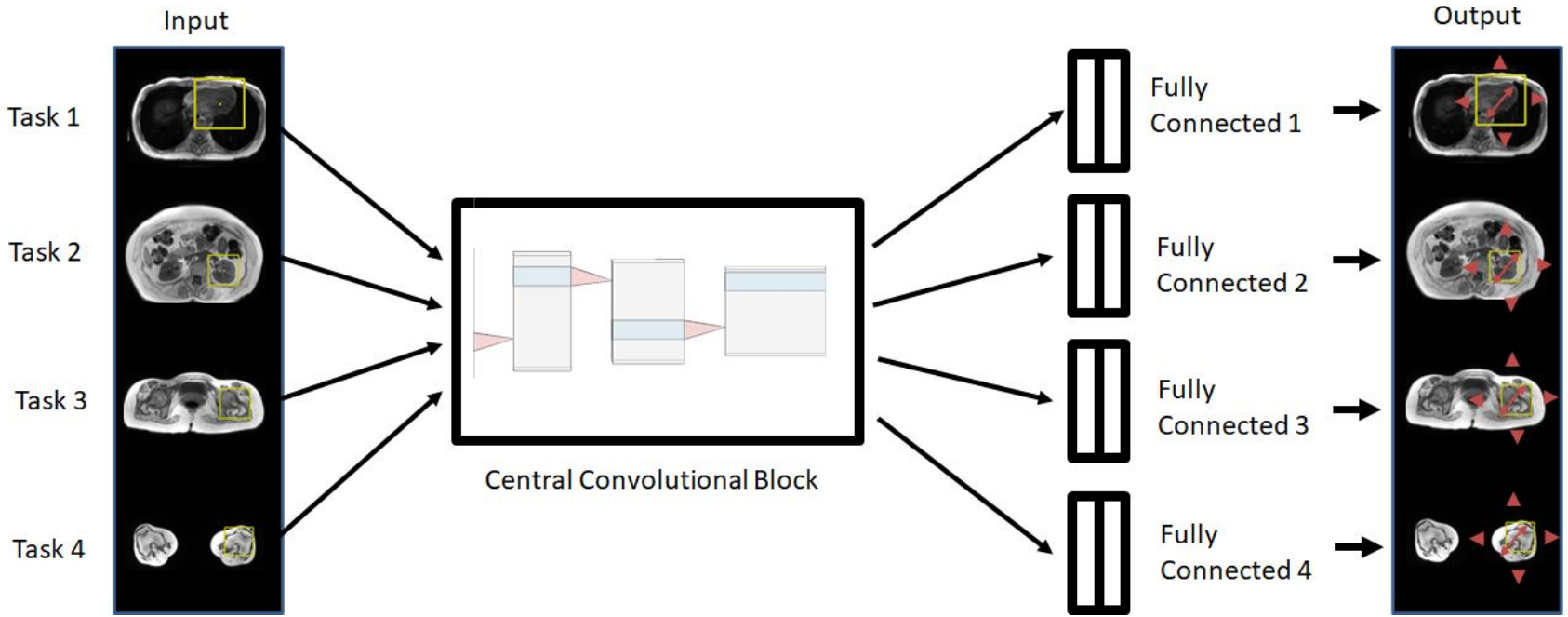
# MARINE System

## Medical image Analysis and Reasoning with Interactive NEtworks



# Lifelong Reinforcement Learning Model

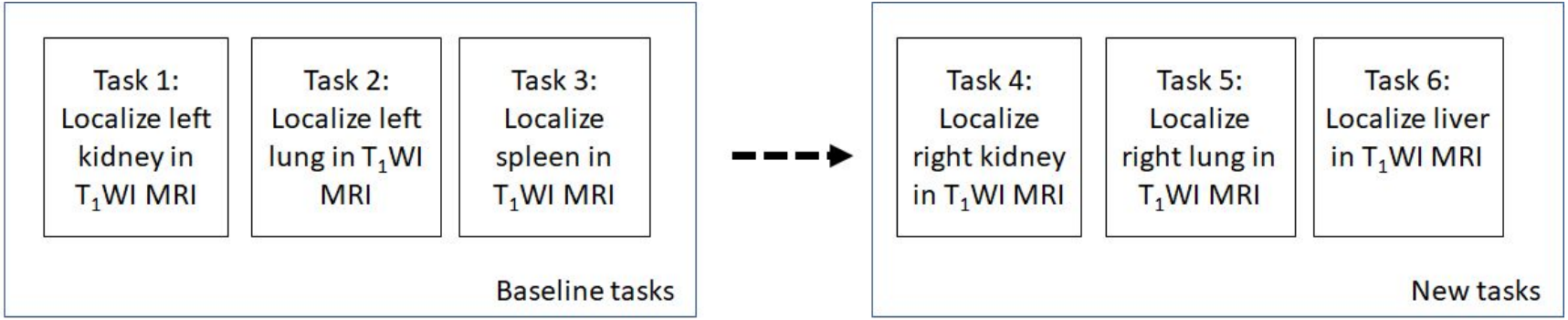
Illustration of a typical 3D DQN trained to localize different anatomical landmarks

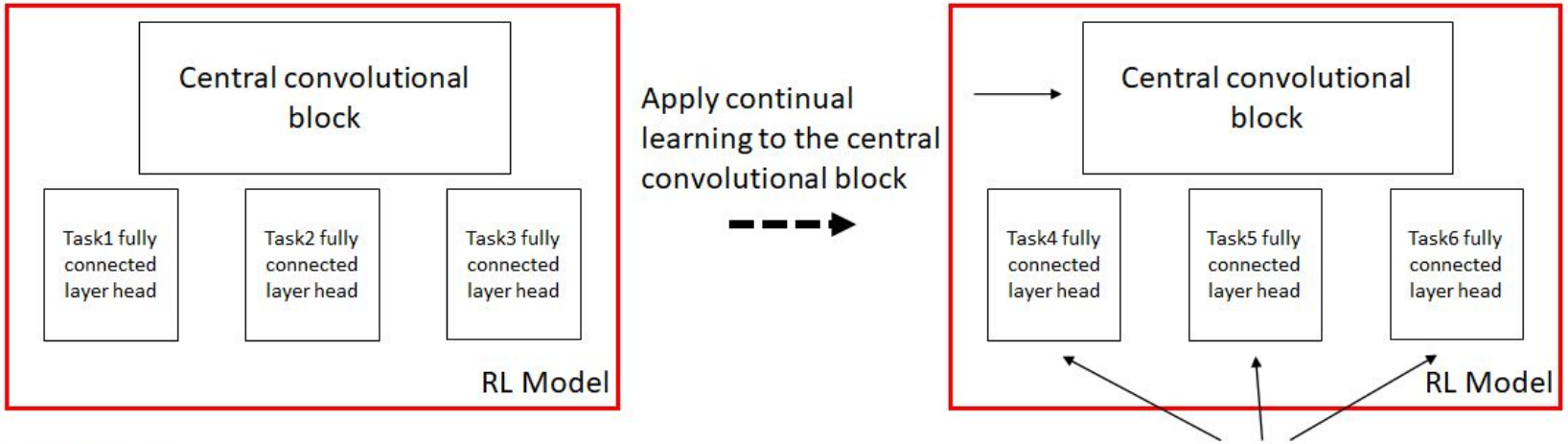



The lifelong reinforcement learning model would need to be updated in three scenarios:

1. New task is added to the database
2. There is an update to an old task (e.g. new patterns)
3. The current node receives updated model(s) from a different nodes across the network.

# Lifelong Reinforcement Learning Model

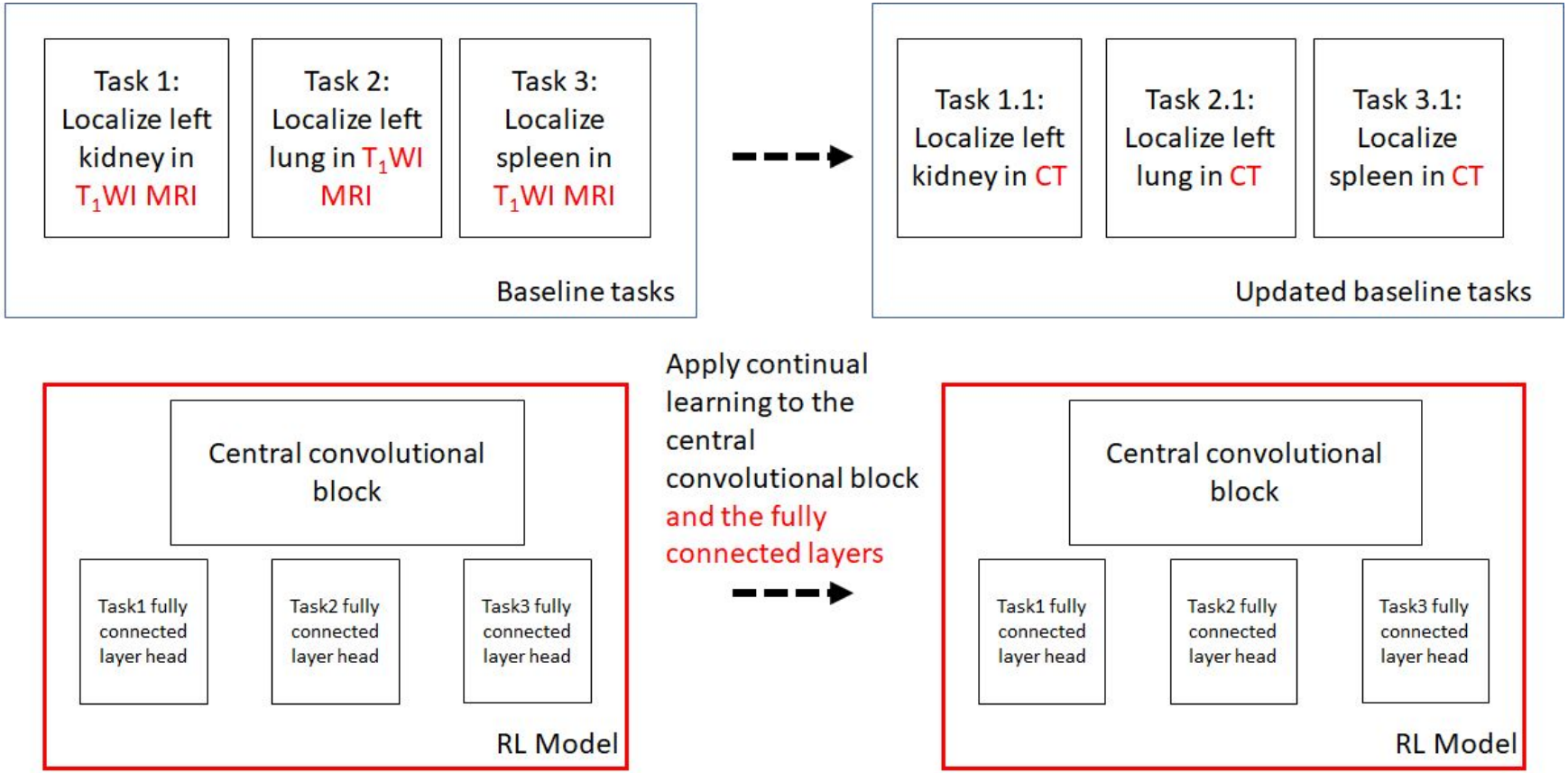



**<u>CASE 1: New Task</u>**

**Task-incremental multi-head continual learning (architecture based)**

# Lifelong Reinforcement Learning Model




**<u>CASE 2: Updated old task</u>**

**Lifelong learning via structural regularization**

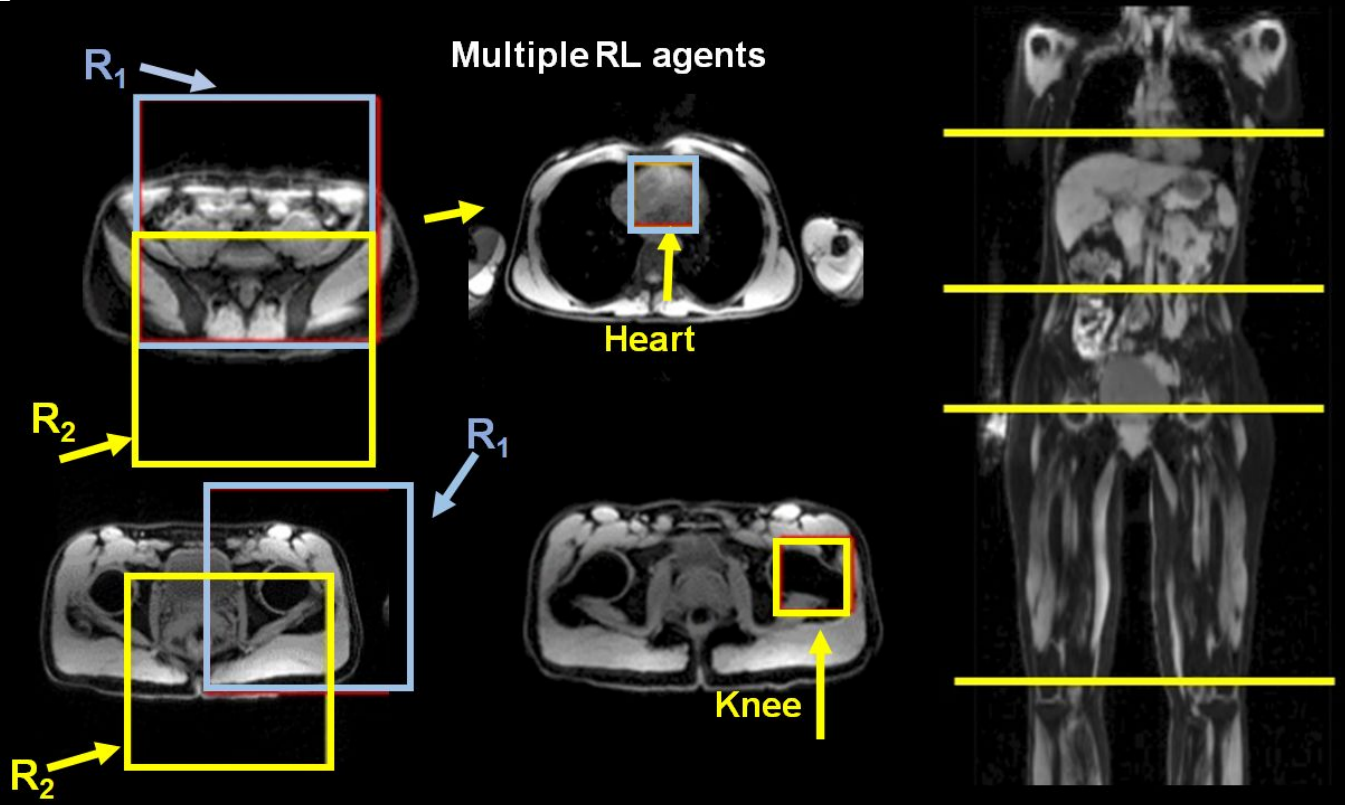
*[This is just a basic idea. We haven't implemented communication yet. Let me know if this sounds like a good idea - Vishwa]*

**Case 3: Model Update**

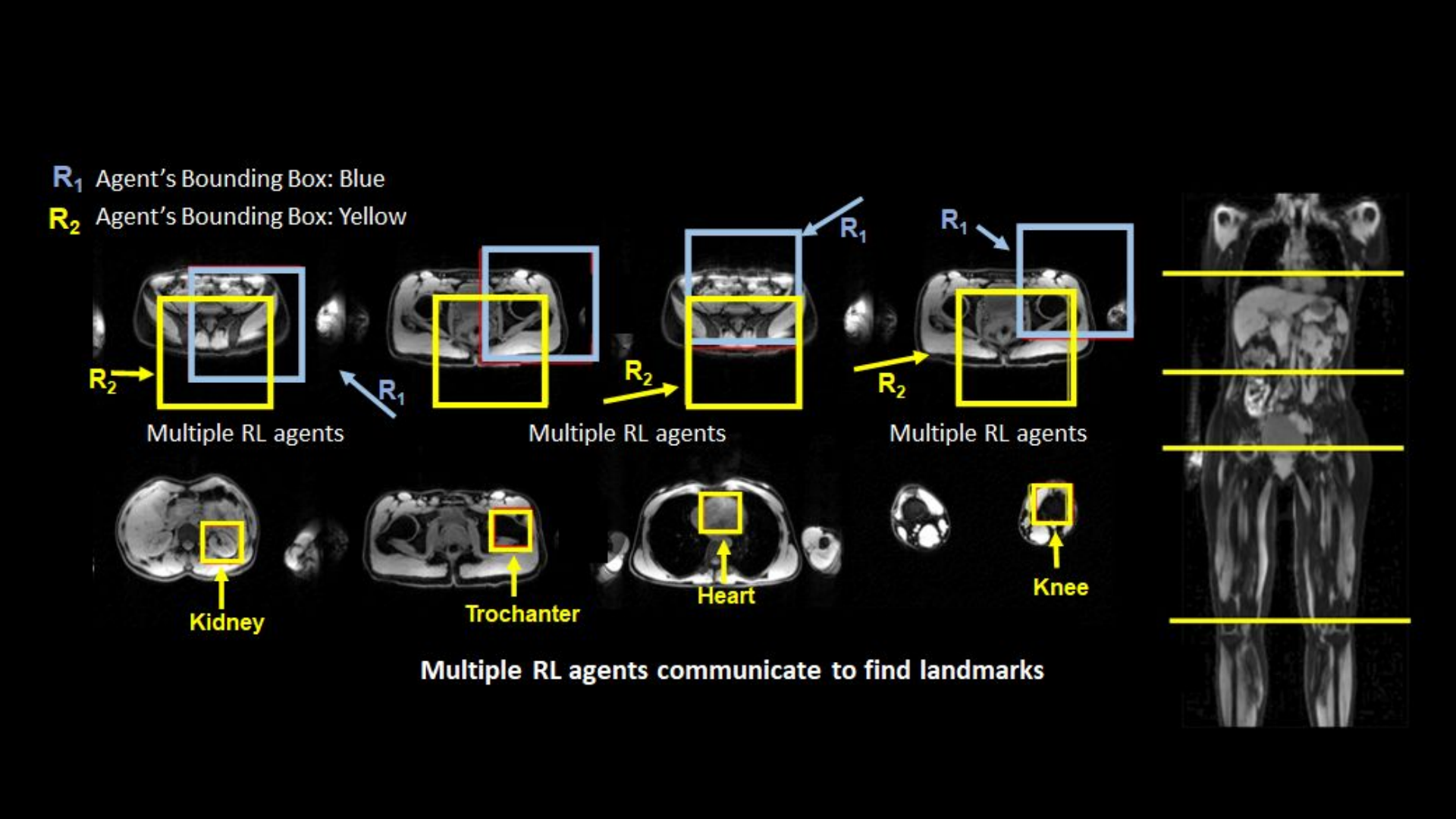
Once the current node receives updated models from different nodes in the network. The following operations would be performed to synchronize the models.

1. Fully connected layers for newer tasks would be directly added to the current network.
2. The weights of the central convolutional block and the fully connected layers (for existing tasks) would be updated by averaging across all the models.

MARINE

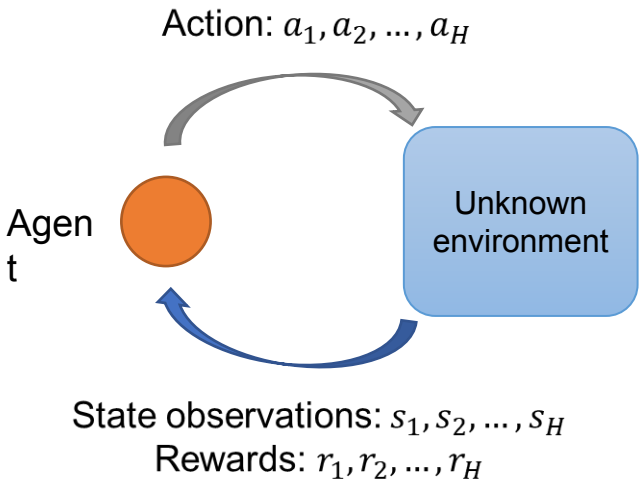



Multiple RL agents communicate to find landmarks


R1 Agent's Bounding Box: Blue
R2 Agent's Bounding Box: Yellow
R2
R1
Multiple RL agents
R1
R2
Multiple RL agents
R1
R2
Multiple RL agents
Kidney
Trochanter
Heart
Knee
Multiple RL agents communicate to find landmarks

# RL with low switching Cost

RL: Learn to interact with an unknown environment through trial and error




Goal: maximize cumulative reward for a horizon $H$

Value: $\mathrm{E}[r_1 + r_2 + r_3 + \cdots r_H]$

**Long term effect** needs to be considered.

# Reinforcement Learning

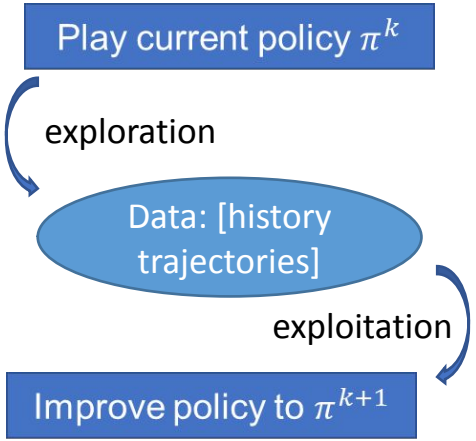
Actions?

States?

Rewards?

# Theories of RL on MDP

- Exploration + exploitation
  - Learn from scratch
  - Exploitation: optimize policy based on existing data
  - Exploration: collect new info about the environment
    - *Regret: average error v.s. optimal policy*



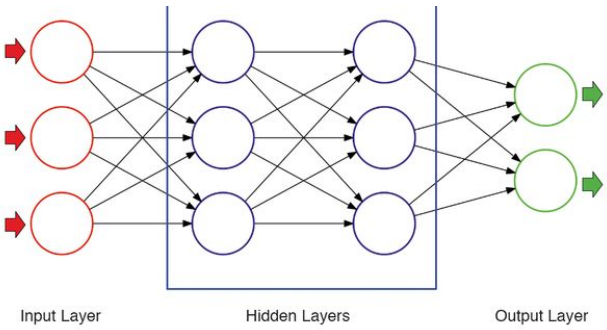

| Algorithm | Ave. Regret | Time | Space |
|---|---|---|---|
| UCRL2 [Jakcsh et al. 2010] | $\geq \tilde{O}(\sqrt{H^4S^2A/T})$ | $\Omega(TS^2A)$ | $O(S^2AH)$ |
| [Agrawal and Jia 2017] | $\geq \tilde{O}(\sqrt{H^3S^2A/T})$ | | |
| UCBVI [Azar et al. 2017] | $\tilde{O}(\sqrt{H^2SA/T})$ | $\tilde{O}(TS^2A)$ | |
| UCB-H [Jin et al. 2018] | $\tilde{O}(\sqrt{H^4SA/T})$ | $O(T)$ | $O(SAH)$ |
| UCB-B [Jin et al. 2018] | $\tilde{O}(\sqrt{H^3SA/T})$ | | |
| Lower bound | $\Omega(\sqrt{H^2SA/T})$ | - | - |

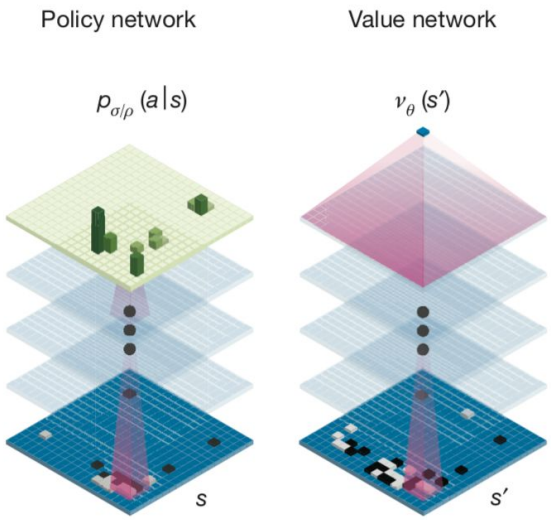
* Recent results include [Zanette&Brunskil' 2018, Zhang et al' 2020 ...]

# General function approximation

- Function class $\mathcal{F}$
  - Might be parametric
  - $f(s,a)$ may rep. $Q^*(s,a)$

- Used in practice




[Silver et al 2016]

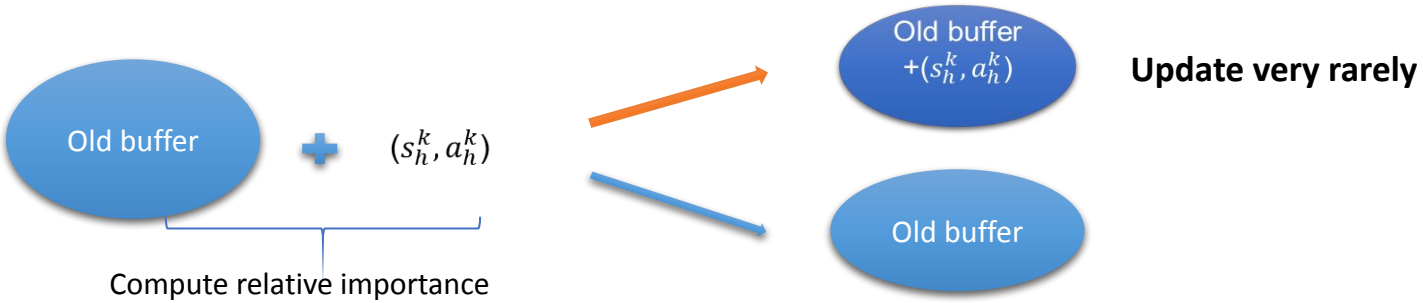
**Issues:**

- Updates policy every round
- Each round takes time to compute

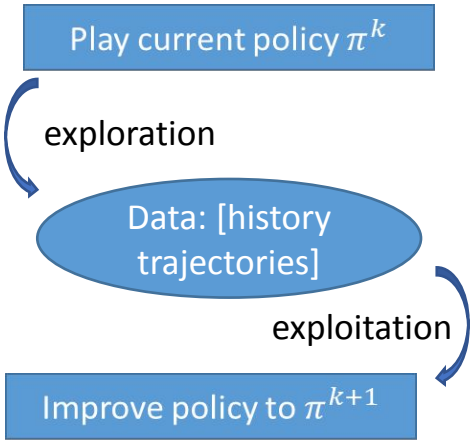
# Improving Efficiency with Online Subsampling

- Needs a full scan over each $t$ points at time $t$
  - Overall running time $\Omega(T^2)$
  - Policy rapidly changing
  - *Can we do better?*

- Online Subsampling
  - Decide to keep or data point upon first seeing
  - Simulates full scan without reading all data



# Low-switching cost RL algorithm




- Policy switches only at
  - $k = 1, 2, 4, 8, 16, 32, \ldots. 2^i$
- For $K = 2^{100}$,
  - Switches only $\sim 100$ times
- Saves computation & communication
  - Full computation only when switches
  - Only to share information when switches